

Notes: Hornbeck (AER, 2012)
**The Enduring Impact of the American Dust Bowl: Short- and Long-Run
Adjustments to Environmental Catastrophe**

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1 Big picture

- **Question.** How did counties affected by the American Dust Bowl adjust in the short run and long run after a large permanent environmental shock?
- **Setting.** The paper studies the 1930s Dust Bowl on the American Plains, where wind and water erosion permanently damaged agricultural land in many counties.
- **Main comparison.** The empirical design compares more-eroded Plains counties to less-eroded Plains counties within the same state and with similar pre-1930 county characteristics.
- **Core outcome.** The paper focuses on agricultural land values, agricultural revenues, land-use choices, population, manufacturing, and local labor-market indicators.
- **Main result.** From 1930 to 1940, the per-acre value of farmland fell by about 30 percent in high-erosion counties and about 17 percent in medium-erosion counties, relative to low-erosion counties.
- **Persistence.** These relative declines did not quickly recover. Land values and agricultural revenues remained persistently lower in more-eroded counties for decades.
- **Limited agricultural adjustment.** Farmers did not rapidly shift farmland away from the activities that erosion made less productive. Adjustment away from cropland and wheat toward pasture and hay was slow and incomplete.
- **Main adjustment margin.** The main margin of adjustment was population decline. More-eroded counties lost population relative to less-eroded counties during the 1930s and continued losing population through the 1950s.
- **Economic takeaway.** Environmental destruction was not mostly offset by local agricultural adaptation. In this setting, long-run economic adjustment occurred mainly through migration and lower local land values rather than rapid changes in local production methods.

2 Incremental contribution of the paper

- **From short-run disaster to long-run adjustment.** The paper is not only about the immediate Dust Bowl shock. It uses a long historical panel to ask how much the local economy eventually adjusted.
- **A historical laboratory for climate and environmental economics.** The Dust Bowl provides a large, spatially varying, and persistent environmental shock. This makes it useful for studying adaptation to permanent productivity losses.
- **Relative adjustment, not just impact.** The paper separates the initial agricultural damage from later adjustment by comparing land values, revenues, land allocations, and population over time.
- **Land values as a forward-looking object.** Because farmland values capitalize expected future rents, they reveal whether farmers and land markets expected productivity losses to be mitigated by future adjustments.

- **A county-level panel contribution.** The paper builds a balanced panel of 779 Plains counties from 1910 to 1997 and combines census agriculture, population, manufacturing, banking, New Deal spending, drought, and erosion data.
- **A mechanism contribution.** The paper studies why agricultural adjustment was slow, including the possible roles of credit constraints, small farm sizes, tenancy, surplus labor, and government programs.
- **A migration-centered adjustment result.** The paper shows that the main response to environmental damage was not local industrialization or local production switching, but large relative population declines.
- **Policy relevance.** The findings warn that adaptation to environmental shocks may be slow and incomplete, especially when capital is scarce and outside labor-market opportunities are weak.

3 Data and measurement

3.1 Dust Bowl erosion measure

- The main erosion variable comes from a Soil Conservation Service map of cumulative erosion damage after the Dust Bowl.
- Counties are classified by the fraction of county area in low, medium, and high erosion regions.
- Low erosion means less than 25 percent of topsoil lost.
- Medium erosion means 25 percent to 75 percent of topsoil lost, possibly with gullies.
- High erosion means more than 75 percent of topsoil lost, possibly with numerous or deep gullies.
- Because detailed erosion surveys began only in response to the Dust Bowl, the paper cannot directly observe pre-1930 erosion. This is a central empirical challenge.

Interpretation: The erosion measure is spatially rich but cumulative. The empirical strategy therefore must address the possibility that some measured erosion reflects pre-existing conditions rather than only the 1930s shock.

3.2 County panel and sample

- The main sample is a balanced panel of 779 Plains counties.
- The sample covers 1910 to 1997, depending on outcome availability.
- County borders are adjusted to maintain 1910 county definitions over time.
- The sample is restricted to a contiguous set of ecologically similar Plains counties using the 1924 USDA *Atlas of Agriculture*.
- The analysis typically weights agricultural regressions by farmland in 1930 and population regressions by population in 1930.

Interpretation: The sample construction makes the comparison more credible by keeping the ecology broadly similar and by preventing county-boundary changes from mechanically driving the results.

3.3 Agricultural outcomes

- **Land values:** value of farmland and buildings per acre of farmland. This is the main forward-looking outcome.
- **Revenues:** value of agricultural products per acre of farmland. This proxies for current agricultural rents.
- **Land-use margins:** farmland share, cropland share, wheat share, and pasture-related measures.
- **Productivity margins:** crop revenue per cropland acre, animal revenue per pasture acre, wheat output per wheat acre, and hay output per hay acre.
- **Agricultural capital:** machinery and equipment values are used in the paper to interpret whether revenue changes overstate or understate rent changes.

Interpretation: The key comparison is between land values and revenues. If revenues fall initially but land values fall much less, markets expect recovery. If both fall similarly and remain low, expected adjustment is limited.

3.4 Population, industry, and local labor-market outcomes

- Population is measured from the decennial Census of Population.
- Manufacturing outcomes include establishments and workers per capita.
- Retail sales per capita proxy for local consumption and earnings when direct wage data are unavailable.
- Unemployment is used to study whether local labor markets cleared after the shock.

Interpretation: These outcomes show whether the local economy adjusted by reallocating workers across sectors locally, or by losing population.

3.5 Potential adjustment barriers

- The paper uses the number of local banks before the Dust Bowl to proxy for access to capital.
- Average farm size proxies for whether farm organization made adjustment easier.
- Tenant share proxies for possible short-horizon land-use incentives.
- Farm population density proxies for surplus agricultural labor.
- New Deal and later government payments are used to examine whether public policy differentially supported or distorted adjustment.

Interpretation: The paper does not merely document persistence. It asks why adaptation was incomplete, with credit constraints receiving the strongest support among the barriers considered.

4 Models and empirical specifications

4.1 Short-run versus long-run agricultural adjustment model

The paper motivates the empirical analysis with a simple agricultural production model. Farmers allocate land between two technologies, for example cropland and pasture or wheat and hay. Before erosion, the farmer chooses land shares so that marginal rents are equalized across uses. The Dust Bowl permanently lowers the productivity of one land use relative to another.

- In the **short run**, land allocations are partly fixed. Rents fall sharply because farmers cannot immediately reallocate land to the more erosion-resistant activity.
- In the **long run**, land can be reallocated. Rents may partly recover if farmers shift land toward less-damaged uses.
- Farmland values fall immediately because they equal the present discounted value of expected future rents.
- If long-run adjustment is expected to be strong, land values should fall less than current revenues.
- If land values fall roughly as much as current revenues, then markets are expecting little recovery in future rents.

Interpretation: Land values are the paper’s key diagnostic tool. They translate expectations about future adaptation into a current price.

4.2 Baseline empirical equation

The main empirical specification compares changes after 1930 across erosion levels:

$$Y_{ct} - Y_{c,1930} = \beta_{1t}M_c + \beta_{2t}H_c + \alpha_{st} + \theta_t X_c + \varepsilon_{ct}. \quad (1)$$

- Y_{ct} is an outcome in county c and year t .
- $Y_{c,1930}$ is the county’s value in 1930, so the dependent variable is the change from the pre-Dust Bowl baseline.
- M_c is the fraction of the county in medium-erosion areas.
- H_c is the fraction of the county in high-erosion areas.
- Low erosion is the omitted category.
- α_{st} are state-by-year fixed effects.
- X_c includes pre-1930 county characteristics and lagged outcomes, each allowed to have year-specific effects.

Interpretation: The coefficients β_{1t} and β_{2t} measure how much more medium- and high-erosion counties changed after 1930 relative to low-erosion counties in the same state, conditional on pre-1930 characteristics.

4.3 Role of control variables

The full specification controls for:

- 1930 county characteristics related to value, land use, population, farms, crop allocation, and animal production.
- available lagged values of these characteristics from 1910, 1920, and 1925.
- the outcome variable in 1930 and all available pre-1930 periods.

Interpretation: The controls allow counties with different pre-Dust Bowl agricultural structures to follow different post-1930 trajectories. This matters because high-erosion counties had different pre-1930 characteristics, such as more corn and animals and less wheat and hay.

4.4 Adjustment-barrier heterogeneity specification

To study whether access to capital or other baseline characteristics affected adjustment, the paper estimates versions of:

$$Y_{ct} - Y_{c,1930} = \beta_{1t}M_c + \beta_{2t}H_c + \beta_{3t}B_c \quad (2)$$

$$+ \beta_{4t}B_c \times M_c + \beta_{5t}B_c \times H_c + \alpha_{st} + \theta_t X_c + \gamma_{1t}L_c + \gamma_{2t}B_c \times L_c + \varepsilon_{ct}. \quad (3)$$

- B_c is a baseline characteristic, such as the log number of banks in 1928.
- B_c is normalized to have standard deviation one.
- β_{4t} and β_{5t} ask whether more-eroded counties adjusted differently when the baseline characteristic was higher.

Interpretation: The most important version uses the number of banks as a proxy for access to capital. If more banks allow more land-use adjustment after erosion, that supports the credit-constraint interpretation.

5 Identification and empirical design

5.1 What the design can identify

- The paper identifies relative changes in more-eroded counties compared with less-eroded counties.
- These estimates are not the same as the aggregate national cost of the Dust Bowl because low-erosion counties may also have been affected indirectly through agricultural prices and migration.
- The design is strongest for measuring differential persistence and adjustment across counties.

5.2 Main identifying assumption

- The key assumption is that, after conditioning on state-by-year effects and pre-1930 county characteristics, more-eroded and less-eroded counties would have followed similar trends absent

Dust Bowl erosion.

- This assumption is plausible only after rich controls because erosion was not randomly assigned.
- Pre-existing geography, agricultural specialization, land values, and pre-1930 erosion could otherwise confound the estimates.

5.3 Why the empirical design is credible

- The paper uses within-state comparisons, absorbing state-level policies, prices, and institutions that vary over time.
- It controls flexibly for pre-1930 county characteristics and lagged outcomes.
- It examines pre-trends, especially in land values before 1930.
- It performs a falsification-style comparison using non-Dust Bowl states with high and medium cumulative erosion, where post-1930 changes should not be driven by Dust Bowl exposure.
- It also uses 1930s weather variation as an instrument for measured erosion in a robustness exercise for land values.

5.4 Remaining limitations

- The erosion measure is cumulative and not a direct measure of erosion that occurred only during the 1930s.
- Weather instruments may violate the exclusion restriction because 1930s weather can affect outcomes directly, not only through erosion.
- The Dust Bowl occurred during the Great Depression, so capital-market and labor-market frictions may have shaped adjustment.
- Relative comparisons may overstate direct out-migration if some migrants moved from high-erosion to low-erosion Plains counties.

Interpretation: The paper is careful: it does not claim a perfect randomized experiment, but it combines rich historical data, flexible controls, pre-trend checks, and robustness analyses to make the relative comparisons informative.

6 Tables and figures

6.1 Figure 1: Aggregate Plains changes and Figure 2: Erosion map

Read:

- Figure 1 shows aggregate changes in farmland acres, population, farmland value, and agricultural revenue for the Plains sample.
- Farmland expanded through the early twentieth century, while agricultural land values and revenues declined during the 1930s and then recovered in aggregate after 1940.

- Figure 2 maps the 779 sample counties and classifies land by low, medium, and high erosion.
- The map shows that the Dust Bowl damage was spatially concentrated but spread across a large part of the Plains.

Economic message: The figures set up the empirical design. Aggregate Plains agriculture eventually recovered, but the map shows that recovery could mask persistent relative losses in more-eroded counties.

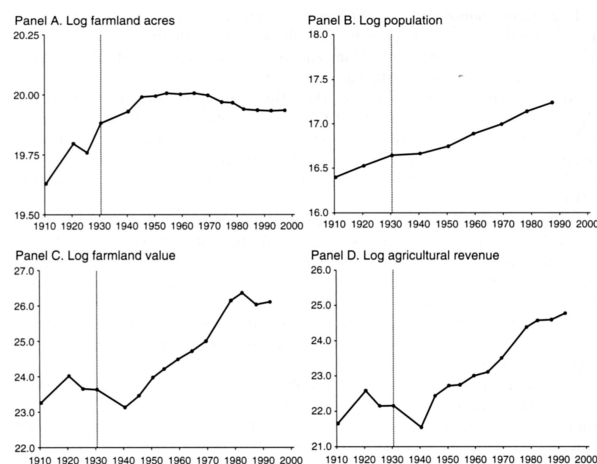


FIGURE 1. AGGREGATE CHANGES ON THE PLAINS IN AGRICULTURE AND POPULATION

Notes: Data are from the US Census of Population and the US Census of Agriculture (Haines 2005; Gutmann 2005). Each panel reports values aggregated in each period over the same 779 sample counties mapped in Figure 2. Panel A shows the natural log of land in acres. Panel B shows the natural log of population. Panel C shows the log of farmland value. Panel D shows the log of agricultural revenue.

(a) Figure 1: Aggregate changes on the Plains

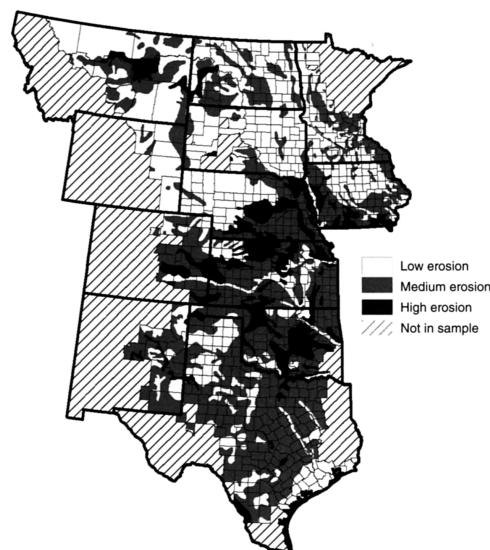


FIGURE 2. THE 779 MAIN SAMPLE COUNTIES, SHADED BY EROSION LEVEL

Notes: Mapped erosion levels are indicated as low (less than 25 percent of topsoil lost), medium (25 percent to 75 percent of topsoil lost and may have some gullies), or high (more than 75 percent of topsoil lost and may have numerous or deep gullies). Thin lines denote 1910 county borders, corresponding to the main sample of 779 counties described in Table 1. Thick lines denote state boundaries. Crossed out areas are not in the sample.

Source: National Archives (College Park, MD), RG 114, Cartographic Records of the Soil Conservation Service, #149.

(b) Figure 2: Main sample counties by erosion level

6.2 Table 1: Baseline county characteristics and Figure 3: Land-value dynamics

Read:

- Table 1 shows that erosion categories were not identical before the Dust Bowl.
- Medium- and high-erosion counties had higher land values and agricultural revenues in 1930 than low-erosion counties.
- High-erosion counties had more corn and animals and less wheat and hay.
- Figure 3 plots estimated differences in log farmland value per acre by erosion level under increasingly rich controls.
- The relative land-value declines appear after 1930 and remain persistent, especially in high-erosion counties.

Economic message: Table 1 explains why controls are necessary. Figure 3 then shows the core reduced-form pattern: after controlling for pre-1930 differences, more-eroded counties experience a large and persistent decline in farmland values.

TABLE 1—COUNTY CHARACTERISTICS IN 1930, BY POST-DUST BOWL EROSION LEVEL

	All counties (1)	Relative to low erosion:		Difference: (3) – (2) (4)
		Medium erosion (2)	High erosion (3)	
<i>Panel A. Value</i>				
Log value of farmland and buildings, per acre in farms	3.377 [0.889]	0.297** (0.094)	0.241 (0.140)	–0.056 (0.102)
Log value of farmland, per acre in farms	3.178 [0.851]	0.264** (0.092)	0.211 (0.136)	–0.052 (0.099)
Log value of all farm products, per acre in farms	1.959 [0.802]	0.268** (0.083)	0.193 (0.122)	–0.074 (0.094)
<i>Panel B. Land use</i>				
Acres of land in farms, per county acre	0.834 [0.153]	0.012 (0.018)	–0.038 (0.021)	–0.050** (0.019)
Acres of cropland, per acre in farms	0.457 [0.236]	0.048 (0.027)	–0.003 (0.036)	–0.050 (0.028)
<i>Panel C. Population and farms</i>				
Population, per 100 county acres	3.242 [6.871]	1.109 (0.694)	1.223 (0.761)	0.114 (0.711)
Fraction of population, rural areas	0.825 [0.215]	–0.014 (0.030)	0.038 (0.042)	0.051 (0.041)
Fraction of population, on farms	0.549 [0.164]	0.035 (0.022)	0.056 (0.031)	0.021 (0.031)
Farms, per 100 county acres	0.292 [0.239]	0.151** (0.023)	0.154** (0.030)	0.003 (0.031)
Average farm size (acres)	685 [1,022]	–341** (128)	–425** (144)	–84 (99)
<i>Panel D. Cropland allocation</i>				
Fraction corn	0.167 [0.163]	0.066** (0.016)	0.193** (0.025)	0.128** (0.023)
Fraction wheat	0.208 [0.217]	–0.054* (0.027)	–0.117** (0.035)	–0.063 (0.036)
Fraction hay	0.130 [0.132]	–0.034 (0.025)	–0.088* (0.038)	–0.054** (0.020)
Fraction cotton	0.108 [0.211]	0.053** (0.019)	0.017 (0.019)	–0.036 (0.021)
Fraction oats, barley, and rye	0.125 [0.113]	0.003 (0.009)	–0.030* (0.012)	–0.033** (0.010)
<i>Panel E. Animal production</i>				
Cattle, per county acre	0.053 [0.033]	0.006 (0.003)	0.010** (0.004)	0.005 (0.004)
Swine, per county acre	0.056 [0.091]	0.035** (0.007)	0.055** (0.012)	0.020 (0.010)
Chickens, per county acre	0.261 [0.272]	0.112** (0.022)	0.121** (0.032)	0.009 (0.031)

Notes: Column 1 reports average values for the 779 sample counties (Figure 2). Counties are weighted by acres of farmland in 1930, and the standard deviation is reported in brackets. Columns 2 and 3 report coefficients from a single regression of the indicated county characteristic in 1930 on the fraction of the county in medium erosion and in high erosion (low erosion is omitted), conditional on state fixed effects and weighted by acres of farmland in 1930. Column 4 reports the difference between the coefficients in columns 2 and 3. Robust standard errors are reported in parentheses.

*Significant at the 1 percent level.

**Significant at the 5 percent level.

(a) Table 1: County characteristics in 1930

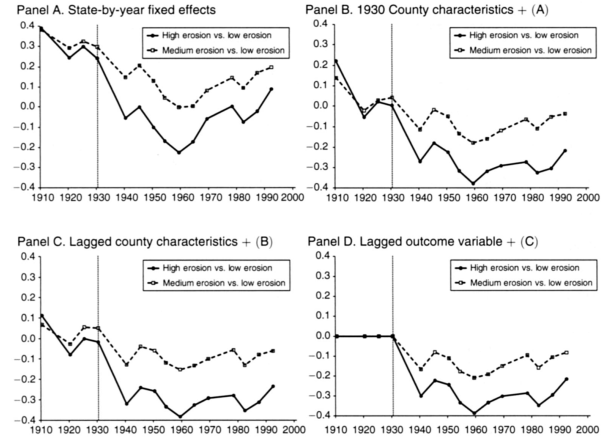


FIGURE 3. ESTIMATED DIFFERENCES IN LOG VALUE OF FARMLAND PER ACRE, BY EROSION LEVEL

Notes: Each panel graphs the estimated coefficients (β) from versions of equation (1) in the text. For each year, the solid circles and solid line report differences in the log per acre value of farmland and buildings for high-erosion counties, relative to low-erosion counties. The hollow squares and dashed line report differences for medium-erosion counties, relative to low-erosion counties. In panel A, these coefficients are estimated by regressing the log per acre value of farmland and buildings on the fraction of a county in a high-erosion area (solid circle) and the fraction of a county in a medium-erosion area (hollow square), controlling for state-by-year fixed effects. Panel B also includes as controls the interaction between each year and each county characteristic in panels B–E of Table 1. Panel C also includes as controls the interaction between each year and the available lagged values of each county characteristic in panels B–E of Table 1 (see Data Appendix). Panel D also includes as controls the interaction between each year and the outcome variable in 1930 and in each lagged year.

(b) Figure 3: Estimated differences in farmland value

6.3 Table 2: Land value and revenue and Table 3: Agricultural production adjustment

Read:

- Table 2 shows that from 1930 to 1940, land values fell by 30 percent in high-erosion counties and 17 percent in medium-erosion counties relative to low-erosion counties.
- Agricultural revenues fell by 27 percent in high-erosion counties and 16 percent in medium-erosion counties relative to low-erosion counties.
- The ratio of land-value declines to revenue declines is close to one, implying that land markets expected little long-run recovery.
- Later panels show that the relative losses in land values and revenues persisted over time.
- Table 3 shows that more-eroded counties had large relative declines in cropland productivity, but only limited and slow reallocation away from cropland.
- Wheat productivity declined in more-eroded counties, but the shift away from wheat toward hay was delayed.

Economic message: These two tables are the heart of the paper. The land-value and revenue evidence implies persistent agricultural costs, while the production table shows that local agricultural adjustment recovered only a small share of the initial damage.

TABLE 2—ESTIMATED CHANGES IN AGRICULTURAL LAND VALUE AND REVENUE, BY EROSION LEVEL

	Change after 1930		Ratio: (1)/(2) (3)
	Log land value (1)	Log revenue (2)	
<i>Panel A. 1940</i>			
High erosion versus low erosion	-0.300** (0.038)	-0.267** (0.055)	1.123** (0.184)
Medium erosion versus low erosion	-0.166** (0.030)	-0.155** (0.040)	1.069** (0.226)
High erosion versus medium erosion (calculated)	-0.134** (0.032)	-0.112* (0.051)	1.197** (0.418)
Averaged value (GLS)			1.109** (0.174)
<i>Panel B. 1945</i>			
High erosion versus low erosion	-0.223** (0.036)	-0.107** (0.038)	
Medium erosion versus low erosion	-0.080** (0.027)	-0.100** (0.030)	
<i>Panel C. 1950–1954 (pooled)</i>			
High erosion versus low erosion	-0.289** (0.041)	-0.240** (0.049)	
Medium erosion versus low erosion	-0.144** (0.032)	-0.187** (0.037)	
<i>Panel D. 1959–1969 (pooled)</i>			
High erosion versus low erosion	-0.340** (0.044)	-0.257** (0.058)	
Medium erosion versus low erosion	-0.183** (0.036)	-0.118** (0.042)	
<i>Panel E. 1978–1992 (pooled)</i>			
High erosion versus low erosion	-0.286** (0.049)	-0.297** (0.080)	
Medium erosion versus low erosion	-0.110** (0.041)	-0.034 (0.060)	
R ²	0.972	0.921	
Sample counties	779	779	

Notes: Columns 1 and 2 report estimates from equation (1) in the text, for the log value of agricultural land and buildings per acre of farmland (column 1) and the log value of agricultural revenue per acre of farmland (column 2). Column 1 corresponds to panel D of Figure 3, where the estimated coefficients are pooled across the indicated years in each panel. Column 3 reports the ratio of the estimated coefficients in panel A of columns 1 and 2. Reported in parentheses are robust standard errors clustered by county.

**Significant at the 1 percent level.
*Significant at the 5 percent level.

TABLE 3—ESTIMATED CHANGES IN AGRICULTURAL PRODUCTION AFTER 1930, BY EROSION LEVEL

	Farmland share (1)	Log cropland productivity (2)	Log pasture productivity (3)	Land share in cropland (4)	Log wheat productivity (5)	Log hay productivity (6)	Land share in wheat (7)
<i>Panel A. High erosion versus low erosion</i>							
1940	-0.025 (0.017)	-0.390** (0.120)	-0.224** (0.044)	-0.000 (0.009)	-0.156 (0.115)		0.035 (0.022)
1945	-0.017 (0.016)	-0.141 (0.074)	-0.116* (0.053)	-0.012 (0.011)	-0.194** (0.070)		
1950–1954	-0.037* (0.016)	-0.382** (0.068)	-0.116* (0.053)	-0.030* (0.014)	-0.230** (0.068)	0.019 (0.047)	0.017 (0.021)
1959–1964	-0.043** (0.016)	-0.333** (0.077)	-0.244** (0.074)	-0.060** (0.017)	-0.214** (0.075)	0.089* (0.045)	-0.001 (0.022)
1969–1974	-0.013 (0.015)			0.021 (0.058)	0.143** (0.044)		-0.039 (0.025)
1978–1992	-0.050** (0.016)				-0.011 (0.045)	0.210** (0.046)	-0.066* (0.028)
1997					0.008 (0.056)	0.139* (0.064)	-0.102** (0.032)
<i>Panel B. Medium erosion versus low erosion</i>							
1940	-0.019 (0.013)	-0.381** (0.089)	-0.060 (0.038)	-0.003 (0.006)	-0.086 (0.085)		-0.008 (0.017)
1945	0.004 (0.012)	-0.289** (0.062)	-0.052 (0.040)	-0.008 (0.007)	-0.064 (0.049)		
1950–1954	-0.008 (0.013)	-0.313** (0.056)	-0.014 (0.041)	-0.018 (0.010)	-0.181** (0.065)	0.017 (0.045)	-0.002 (0.020)
1959–1964	-0.025 (0.014)	-0.275** (0.068)	0.006 (0.054)	-0.028* (0.012)	-0.257** (0.062)	0.041 (0.046)	-0.026 (0.021)
1969–1974	-0.011 (0.012)				-0.039 (0.044)	-0.028 (0.036)	-0.070** (0.023)
1978–1992	-0.026* (0.013)				-0.064* (0.033)	-0.004 (0.039)	-0.110** (0.023)
1997					-0.066 (0.050)	-0.016 (0.050)	-0.138** (0.025)
R ²	0.656	0.791	0.850	0.679	0.890	0.823	0.435
Sample counties	779	779	779	779	421	421	421
Weighted by 1930 value of:	Farmland	Cropland	Pasture	(2) + (3)	Wheat	Hay	(5) + (6)

Notes: Each column reports estimates from equation (1) in the text: farmland per county acre (column 1), log crop revenue per cropland acre (column 2), log animal revenue per pasture acre (column 3), cropland per acre of cropland and pasture (column 4), log wheat output per wheat acre (column 5), log hay output per hay acre (column 6), wheat per acre of wheat and hay (column 7). Reported in parentheses are robust standard errors clustered by county.

**Significant at the 1 percent level.
*Significant at the 5 percent level.

(a) Table 2: Agricultural land value and revenue

(b) Table 3: Agricultural production after 1930

6.4 Table 4: Population and manufacturing and Table 5: Barriers to land-use adjustment

Read:

- Table 4 shows that population fell sharply in more-eroded counties after the Dust Bowl.
- By 1940, population had fallen by 11.6 percent in high-erosion counties and 8.8 percent in medium-erosion counties relative to low-erosion counties.
- Relative population losses persisted and grew through 1960.
- Manufacturing did not expand enough to absorb displaced labor; there is no strong evidence of compensating local industrialization.
- Retail sales per capita fell in 1940, suggesting lower local consumption and earnings in more-eroded counties.
- Table 5 studies whether adjustment was faster in counties with more banks, larger farms, less tenancy, or less dense farm population.
- Counties with more pre-Dust Bowl banks shifted more land from cropland to pasture, which supports a role for credit constraints.

Economic message: Local labor-market adjustment occurred mainly through migration rather than industrial growth. The heterogeneity results suggest that access to finance helped land-use adjustment, but other barriers receive weaker support.

TABLE 4—ESTIMATED CHANGES IN POPULATION AND MANUFACTURING AFTER 1930, BY EROSION LEVEL

	Log population (1)	Log mfg. establishments (2)	Mfg. workers per capita (3)	Unemployment rate (4)	Log retail sales per capita (5)
<i>Panel A. High erosion versus low erosion</i>					
1940	-0.116** (0.021)	-0.015 (0.069)	0.003 (0.002)	0.010** (0.003)	-0.122** (0.029)
1950	-0.180** (0.047)	-0.183** (0.067)		-0.004 (0.002)	
1960	-0.267** (0.072)	-0.188* (0.082)	0.005 (0.007)		-0.066* (0.029)
1970	-0.268** (0.088)	-0.104 (0.100)	0.006 (0.010)		-0.044 (0.037)
1980	-0.250* (0.101)	-0.041 (0.117)	0.013 (0.013)		-0.093* (0.043)
1990	-0.227* (0.114)	-0.087 (0.136)	0.010 (0.012)		-0.090 (0.051)
<i>Panel B. Medium erosion versus low erosion</i>					
1940	-0.088** (0.017)	-0.100* (0.047)	-0.001 (0.002)	0.003 (0.002)	-0.066** (0.021)
1950	-0.145** (0.033)	-0.084 (0.046)		-0.002 (0.002)	
1960	-0.236** (0.054)	-0.139* (0.062)	0.005 (0.005)		0.023 (0.021)
1970	-0.246** (0.070)	-0.116 (0.075)	0.012 (0.008)		0.041 (0.026)
1980	-0.251** (0.082)	-0.036 (0.092)	0.018* (0.009)		-0.021 (0.032)
1990	-0.223* (0.094)	-0.026 (0.107)	0.024** (0.008)		-0.013 (0.039)
R ²	0.654	0.637	0.612	0.866	0.998
Sample counties	779	516	287	779	758

Notes: Each column reports estimates from equation (1) in the text for the indicated outcome variable. In columns 2, 3, and 5, manufacturing and retail data are combined with population data from the nearest decade and the estimated coefficients are pooled over the following years: 1945 and 1954 for 1950; 1959 and 1964 for 1960; 1969 and 1974 for 1970; 1978 and 1982 for 1980; 1987 and 1992 for 1990. All regressions are weighted by county population in 1930. Reported in parentheses are robust standard errors clustered by county.

**Significant at the 1 percent level.

*Significant at the 5 percent level.

TABLE 5—ESTIMATED CHANGES IN AGRICULTURAL LAND USE AFTER 1930, INTERACTED WITH COUNTY PRECHARACTERISTICS

Relative adjustment in areas with one standard deviation:							
More banks		Larger farms		Less tenant farming		Less dense farm population	
Land share in cropland (1)	Land share in wheat (2)	Land share in cropland (3)	Land share in wheat (4)	Land share in cropland (5)	Land share in wheat (6)	Land share in cropland (7)	Land share in wheat (8)
<i>Panel A. High erosion versus low erosion</i>							
1940	-0.043** (0.015)	0.026 (0.019)	0.025* (0.011)	-0.010 (0.017)	0.018 (0.009)	0.008 (0.017)	0.030** (0.011)
1945	-0.038* (0.017)	0.014 (0.014)	0.006 (0.011)	0.006 (0.011)	0.006 (0.011)	0.006 (0.011)	0.021 (0.014)
1950–1954	-0.056** (0.019)	0.007 (0.021)	0.034* (0.017)	-0.034 (0.020)	0.019 (0.015)	-0.057** (0.019)	0.040* (0.016)
1959–1964	-0.058** (0.021)	0.016 (0.022)	0.027 (0.020)	-0.008 (0.023)	0.018 (0.017)	-0.033 (0.023)	0.037 (0.019)
<i>Panel B. Medium erosion versus low erosion</i>							
1940	-0.019** (0.006)	-0.010 (0.012)	0.010 (0.007)	0.020 (0.014)	0.009 (0.007)	0.012 (0.013)	0.013 (0.007)
1945	-0.019** (0.007)	0.004 (0.008)	0.004 (0.008)	-0.005 (0.008)	-0.005 (0.008)	-0.005 (0.008)	0.005 (0.008)
1950–1954	-0.011 (0.009)	-0.002 (0.017)	0.000 (0.012)	-0.027 (0.018)	-0.015 (0.012)	-0.052** (0.016)	0.002 (0.012)
1959–1964	-0.015 (0.011)	0.002 (0.017)	0.006 (0.015)	-0.021 (0.019)	-0.020 (0.016)	-0.055** (0.017)	0.007 (0.015)
R ²	0.695	0.468	0.688	0.472	0.689	0.491	0.689
Sample counties	705	705	705	705	705	705	705
Weighted by 1930:	Cropland + Pasture	Wheat + Hay	Cropland + Pasture	Wheat + Hay	Cropland + Pasture	Wheat + Hay	Cropland + Pasture

Notes: Each column reports estimates from equation (2) in the text for the indicated outcome variable. For columns 1 and 2, the interaction term is the log number of banks at the end of 1928. For columns 3 and 4, the interaction term is the log average farm size in 1930. For columns 5 and 6, the interaction term is the tenant share of farmland in 1930. For columns 7 and 8 the interaction term is the log farm population per acre of farmland in 1930. All interaction terms are normalized to have mean zero and standard deviation of one (negative one for columns 5–8). Reported in parentheses are robust standard errors clustered by county.

**Significant at the 1 percent level.

*Significant at the 5 percent level.

(a) Table 4: Population and manufacturing after 1930

(b) Table 5: Land-use adjustment and precharacteristics

6.5 Table 6: Government payments and Figure A1: Dust Bowl versus non-Dust Bowl states

Read:

- Table 6 reports New Deal payments in the 1930s and Conservation Reserve Program payments in 1992.
- There is little evidence that more-eroded counties received systematically greater 1930s New Deal payments per farm acre.
- In 1992, more-eroded counties received more Conservation Reserve Program payments and a higher fraction of payments through CRP.
- Figure A1 compares erosion-related land-value patterns in Dust Bowl states with analogous erosion categories in non-Dust Bowl states.
- The sharp post-1930 relative decline appears in the Dust Bowl states, not in the same way in non-Dust Bowl states.

Economic message: Government programs do not explain the main 1930s relative decline. The non-Dust Bowl comparison strengthens the interpretation that the post-1930 land-value pattern reflects Dust Bowl erosion rather than generic differences among high- and low-erosion counties.

TABLE 6—ESTIMATED DIFFERENCES IN GOVERNMENT PAYMENTS PER FARM ACRE, BY EROSION LEVEL

	All Counties (1)	Relative to low erosion:		Difference: (3) – (2) (4)
		Medium Erosion (2)	High Erosion (3)	
<i>Panel A. New Deal payments (1933–39)</i>				
AAA payments	0.459 [0.325]	0.012 (0.016)	–0.030 (0.022)	–0.042* (0.021)
Public works spending	0.229 [0.568]	0.019 (0.049)	–0.055 (0.060)	–0.074 (0.061)
Relief spending	0.426 [2.167]	0.099 (0.075)	0.167 (0.107)	0.068 (0.091)
New Deal loans	0.446 [1.094]	–0.058 (0.080)	–0.037 (0.095)	0.021 (0.080)
Mortgage loans guaranteed	0.112 [0.801]	0.018 (0.033)	–0.067 (0.050)	–0.085* (0.039)
<i>Panel B. Government payments in 1992</i>				
Conservation Reserve Program	1.430 [1.298]	0.713** (0.140)	1.333** (0.226)	0.619** (0.211)
All government payments	6.659 [5.206]	–0.666 (0.408)	–1.655** (0.507)	–0.989* (0.464)
Fraction of payments through CRP	0.248 [0.158]	0.065** (0.020)	0.162** (0.027)	0.096** (0.024)

Notes: Panel A reports differences in 1930's New Deal spending across 776 counties. Panel B reports differences in conservation reserve program (CRP) payments in 1992, all government payments in 1992, and the fraction of payments through the CRP in 1992. Column 1 reports the mean and standard deviation in brackets. Column 2 (column 3) reports the average difference for medium-erosion counties (high-erosion counties) relative to low-erosion counties, controlling for state fixed effects and county characteristics in panels B–E of Table 1, and lagged values of the characteristics. Column 4 reports the difference in coefficients between columns 2 and 3. The means and regressions are weighted by county farmland in 1930. Robust standard errors are reported in parentheses.

** Significant at the 1 percent level.

* Significant at the 5 percent level.

(a) Table 6: Government payments by erosion level

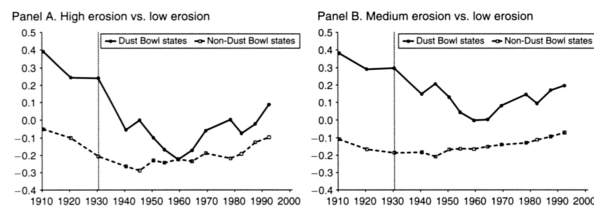


FIGURE A1. ESTIMATED DIFFERENCES BY EROSION LEVEL IN LOG VALUE OF FARMLAND: DUST BOWL VS. NON-DUST BOWL STATES

Notes: The solid line in both panels corresponds to estimated within-state differences by erosion level in the log value of agricultural land and buildings per acre of farmland (from panel A of Figure 3), for the 779 sample counties in Figure 2. The dashed line in both panels corresponds to estimated within-state differences by erosion level for 1,848 counties in other US states (not those depicted in Figure 2). Panel A reports the difference between high-erosion and medium-erosion counties; panel B reports the difference between medium-erosion and low-erosion counties.

(b) Figure A1: Dust Bowl versus non-Dust Bowl states

6.6 Figure A2: Local banks and Table A1: Weather instruments

Read:

- Figure A2 shows relative changes in the log number of local banks at year end by erosion level.
- High-erosion counties experienced a sharp relative loss of local banks in the early 1930s, consistent with a decline in local credit access.
- Table A1 uses 1930s drought and temperature variables to predict high and medium erosion and then instruments for erosion in land-value regressions.
- The first stage shows that extreme drought and average drought severity predict high erosion.
- The 2SLS estimate for high-erosion counties is more negative than the OLS estimate, though the author emphasizes that weather may have direct effects and should be interpreted cautiously.

Economic message: The appendix evidence supports two important points: credit access worsened in the most damaged counties, and measured erosion is meaningfully related to 1930s weather shocks. Both help interpret the main results, while also highlighting the empirical challenges.

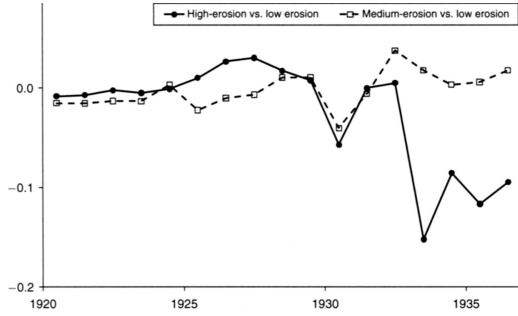


FIGURE A2. ESTIMATED DIFFERENCES IN LOG NUMBER OF LOCAL BANKS AT YEAR END, BY EROSION LEVEL

: This figure graphs the estimated coefficients (β) from a modified version of equation (1), described in panel figure 3. Bank data are available annually, so the regression controls for state-by-year fixed effects and county characteristics interacted with every year.

(a) Figure A2: Local banks by erosion level

AND INSTRUMENTAL VARIABLES ESTIMATES FOR CHANGES IN LAND VALUES FROM 1930 TO 1940

	First stage:		Change in land value, 1930–1940	
	Fraction high-erosion (1)	Fraction medium-erosion (2)	2SLS (3)	OLS (4)
<i>Panel A. Erosion level</i>				
High-erosion versus low-erosion			−0.996** (0.229)	−0.312** (0.037)
Medium-erosion versus low-erosion			−0.137 (0.175)	−0.175** (0.031)
<i>Panel B. Weather instruments</i>				
Months in extreme drought, 1930s	0.0070** (0.0018)	0.0056* (0.0028)		
Months in severe drought, 1930s	0.0020 (0.0015)	0.0085** (0.0023)		
Months in moderate drought, 1930s	0.0001 (0.0016)	0.0038 (0.0024)		
Average Palmer Drought Severity, 1930s	0.0897** (0.0313)	0.1640** (0.0578)		
Average annual temperature, 1930s	−0.0277 (0.0254)	0.0957* (0.0413)		
Temperature standard deviation, 1930s	0.1209* (0.0518)	0.1424 (0.0863)		
F-stat: Instruments	8.376	52.77		
p-value: Instruments = 0	0.0002	0.0000		
Sample counties	776	776	776	776

Notes: Column 1 reports first-stage estimates from regressing the fraction of a county in high-erosion areas on the 1930s weather instruments, controlling for the same weather variables from 1895–1929 and the county controls included in Table 2. Column 2 reports analogous estimates for the fraction of a county in medium-erosion areas. Column 3 reports two-stage least squares estimates of the relative change in land values by erosion level, controlling for the variables in columns 1 and 2 and instrumenting for erosion levels with the 1930s weather instruments. Column 4 reports OLS estimates (as in Table 2), with the same controls and sample as column 3. Robust standard errors are reported in parentheses.

** Significant at the 1 percent level.

* Significant at the 5 percent level.

(b) Table A1: Weather instruments and IV estimates

7 Short summary

- The paper studies the long-run economic consequences of the American Dust Bowl.
- The core sample is a balanced panel of 779 Plains counties from 1910 to 1997.
- Counties are compared by cumulative erosion severity: low, medium, and high erosion.
- The main empirical design estimates changes after 1930 in more-eroded counties relative to less-eroded counties within the same state and with similar pre-1930 characteristics.
- The main result is a large and persistent decline in farmland values and agricultural revenues in more-eroded counties.
- From 1930 to 1940, farmland values fell by about 30 percent in high-erosion counties and 17 percent in medium-erosion counties relative to low-erosion counties.
- Agricultural revenues fell by similar magnitudes, and later recovery was limited.
- Land-use adjustment was slow: farmers did not quickly move away from cropland and wheat toward pasture and hay.
- The main margin of adjustment was population decline: more-eroded counties lost population during the 1930s and through the 1950s.
- There is little evidence that manufacturing expanded enough to absorb displaced labor locally.
- Credit constraints appear to have limited adjustment: more banks before the 1930s predict faster crop-to-pasture adjustment.
- The paper's broader lesson is that adaptation to environmental damage may be incomplete and may require large geographic reallocation of people.

8 Conclusion

8.1 Core conclusion

Hornbeck (2012) shows that the American Dust Bowl imposed large, persistent, and spatially uneven agricultural costs. More-eroded counties did not mostly recover through rapid local agricultural adaptation. Instead, adjustment occurred mainly through lower farmland values and large relative population declines.

8.2 Economic intuition

The Dust Bowl permanently reduced the productivity of damaged soils. If farmers could quickly reallocate land to less erosion-sensitive uses, then current revenues would fall but land values would fall less because future rents would recover. Instead, land values and revenues fell by similar magnitudes, showing that markets expected limited recovery. The slow land-use response implies that productive opportunities to adapt existed but were hard to implement quickly.

8.3 Incremental contribution revisited

The paper contributes to environmental economics by showing that long-run adaptation can be slow and incomplete. It contributes to economic history by measuring the local economic impact of the Dust Bowl with county-level data. It contributes to regional economics by showing that labor-market adjustment to a negative local shock occurred mainly through migration rather than local sectoral transformation.

8.4 Implications

- Environmental shocks can have persistent local economic effects even when agents have decades to adjust.
- Land markets can reveal expectations about future adaptation because land values capitalize future rents.
- Migration may be a central part of adaptation to environmental damage.
- Credit-market frictions can slow the adoption of new land-use patterns.
- Policy evaluations of climate change and environmental degradation should not assume that local production adjustments will fully offset damage.

8.5 Limitations

- The analysis estimates relative effects across Plains counties, not the full national aggregate cost.
- The erosion measure is cumulative and may include pre-Dust Bowl erosion.
- The Dust Bowl coincided with the Great Depression, which may have limited capital access and outside employment opportunities.

- Weather-based IV estimates require strong assumptions because drought and temperature can directly affect outcomes.
- The paper cannot fully observe all private adjustment costs, expectations, or migration destinations.

8.6 Takeaway

The paper's main message is that the economic impact of environmental catastrophe can endure for decades. In the Dust Bowl, local agricultural adaptation was limited, and the economy adjusted primarily by moving people out of the most damaged places.